Lab Manual - LAB 2

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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INSTITUTE OF MANAGEMENT SCIENCES, PESHAWAR

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LAB 2: Cisco Packet Tracer Setup and Basic Network

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Cisco Packet Tracer Installation and Interface

Learning Objective

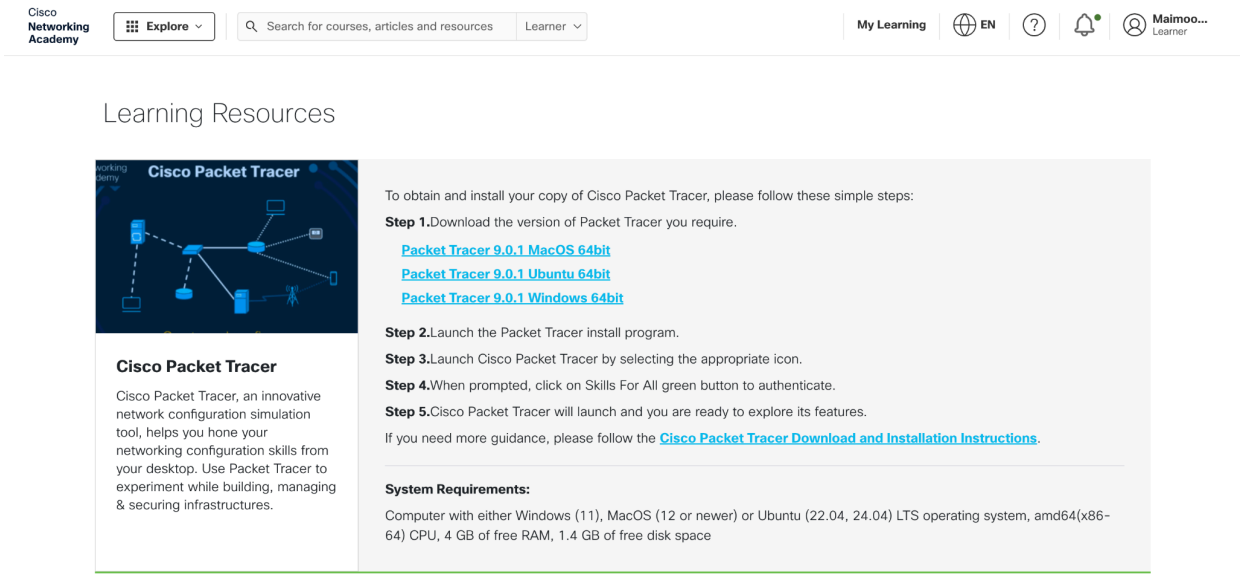
By the end of this lab, students will be able to:

- Install and launch Cisco Packet Tracer.
- Create and save a Packet Tracer activity file.
- Identify the major areas of the Packet Tracer interface.
- Locate network devices, end devices, connections, and workspace tools.

Install Cisco Packet Tracer

Step 1: Download Cisco Packet Tracer

Download and install Cisco Packet Tracer from the official Cisco Networking Academy platform. https://www.netacad.com/resources/lab-downloads?courseLang=en-US



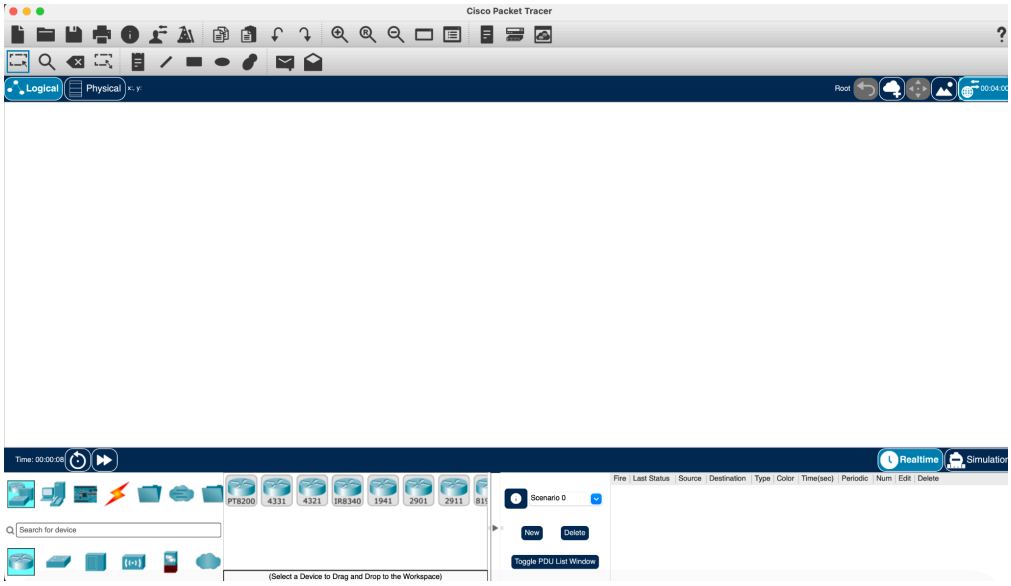
Step 2: Install Packet Tracer

Run the downloaded installer and follow the installation wizard. Accept the license agreement and keep the default installation settings unless instructed otherwise.

Step 3: Launch Packet Tracer

After installation, open Cisco Packet Tracer.

If prompted to sign in, use your Cisco Networking Academy account.



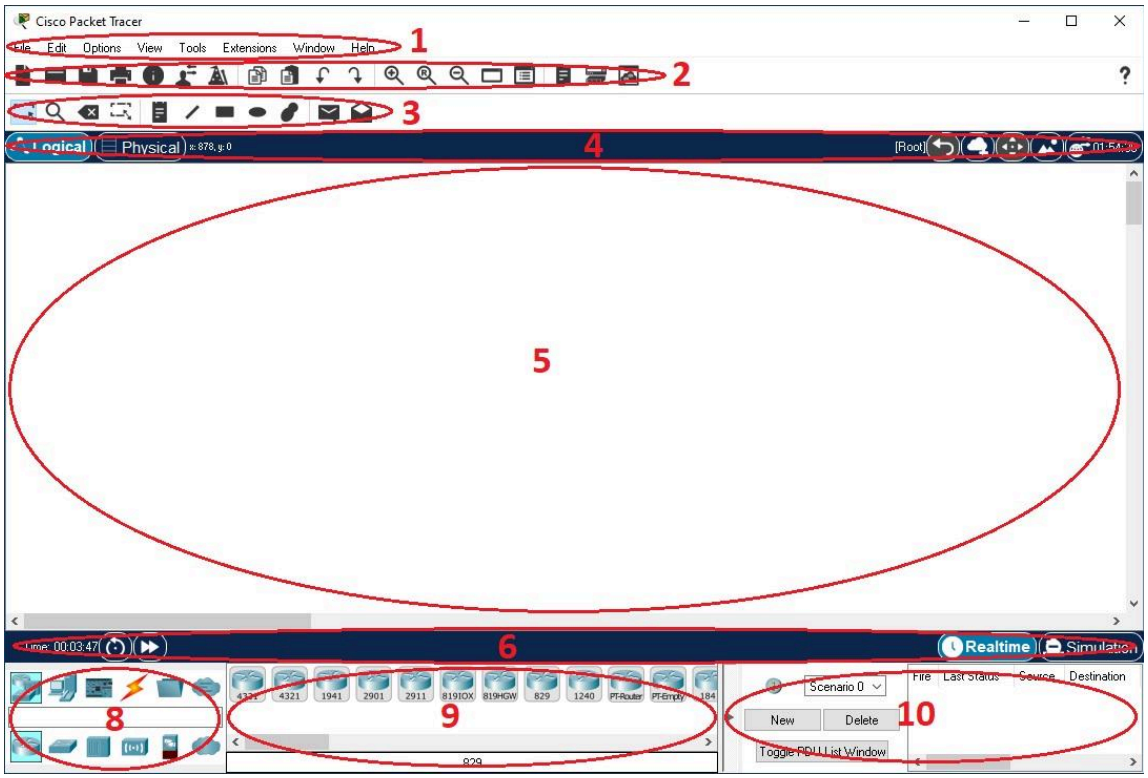
Explore the Packet Tracer Interface

When Packet Tracer opens, students will see a graphical workspace where network devices can be placed, connected, configured, and tested.

	Interface	Description
1	**Menu Bar**	This bar provides the **File, Edit, Options, View, Tools, Extensions, Window** and **Help menus**. You will find basic commands such as Open, Save, Save as Pkz, Print, and Settings and Preferences in these menus. You will also be able to access the Activity Wizard from the Extensions menu. Window menu

		allows to enter/exit full screen mode, lists all top level windows open by Packet Tracer and allows to activate them.
2	**Main Tool Bar**	This bar provides shortcut icons to most commonly used menu commands.
3	**Common Tools Bar**	This bar provides access to these commonly used workspace tools: **Select**, **Inspect**, **Delete**, **Resize Shape**, **Place Note**, **Drawing Palette**, **Add Simple PDU(protocol data unit)**, and **Add Complex PDU**. See "Workspace Basics" for more information.
4	**Logical/Physical Workspace and Navigation Bar**	You can toggle between the Physical Workspace and the Logical Workspace with the tabs on this bar. In Logical Workspace, this bar also allows you to go back to a previous level in a cluster, create a **New Cluster**, **Move Object**, **Set Tiled Background**, and **Viewport**. In Physical Workspace, this bar allows you to navigate through physical locations, create a **New City**, create a **New Building**, create a **New Closet**, **Move Object**, apply a **Grid** to the background, **Set Background**, and go to the **Working Closet**.
5	**Workspace**	This area is where you will create your network, watch simulations, and view many kinds of information and statistics.
6	**Realtime/Simulation Bar**	You can toggle between Realtime Mode and Simulation Mode with the tabs on this bar. This bar also provides buttons to **Power Cycle Devices** and **Fast Forward Time** as well as the **Play Control** buttons and the **Event List** toggle button in Simulation Mode. Also, it contains a clock that displays the relative **Time** in Realtime Mode and Simulation Mode.
7	**Network Component Box**	This box is where you choose devices and connections to put into the workspace. It contains the **Device-Type Selection** Box and the **Device-Specific Selection** Box. There's a searchable field that allows you to enter a device name to look for that specific device quickly. Device name shows when you mouse over the device icon in the Device-Specific Box.

8	**Device-Type Selection Box**	This box contains the type of devices and connections available in Packet Tracer. The **Device-Specific Selection** Box will change depending on which type of device you choose.
9	**Device-Specific Selection Box**	This box is where you choose specifically which devices you want to put in your network and which connections to make. In this box, you'll find devices that may have already been obsolete. You have an option to hide legacy equipment in the Preferences window under Options.
10	**User Created Packet Window**	This window manages the packets you put in the network during simulation scenarios. See the "Simulation Mode" section for more details.

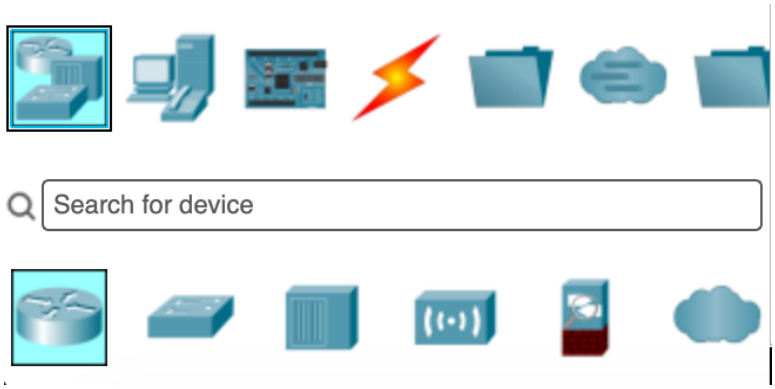


Step 1: Explore Network Devices

From the device selection area, locate:

- Routers
- Switches
- Hubs
- Wireless devices

Click on the different categories and observe the available devices.



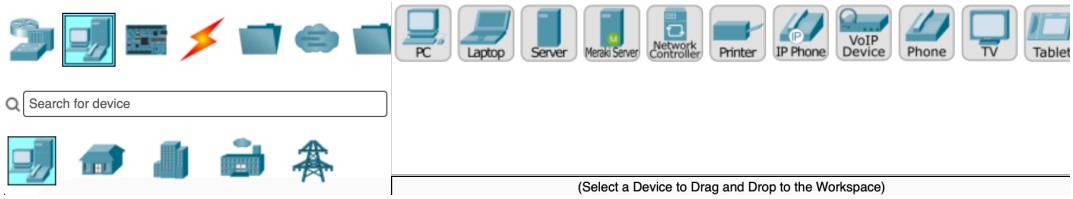
Step 2: Explore End Devices

Select the **End Devices** category.

Locate:

- PC
- Laptop
- Server
- Printer

These devices will be used throughout the laboratory exercises.

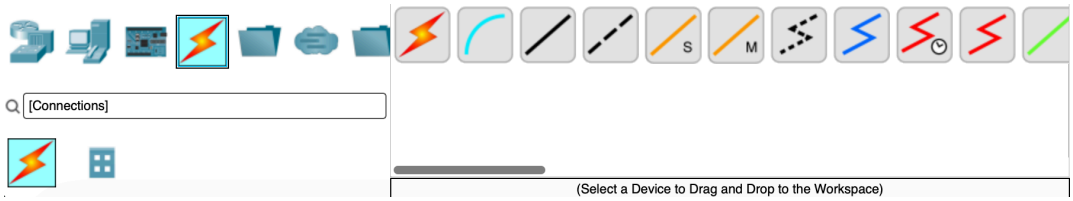


Step 3: Explore Connections

Select the **Connections** category. Observe the available cable types, including:

- Copper Straight-Through
- Copper Cross-Over
- Fiber
- Console
- Serial

For this experiment, we will primarily use the **Copper Straight-Through** cable.



Step 4: Save a Packet Tracer File

Create a new Packet Tracer file and save it using the following format:

File Name:

Week1_Experiment1_YourName.pkt

Realtime and Simulation Modes

Objective

Students will learn how to switch between **Realtime** and **Simulation** modes and observe network communication.

Realtime Mode

Realtime mode is the normal operating mode of Packet Tracer. In this mode, network devices behave approximately like real devices. When a packet is sent, communication takes place automatically without showing every individual step.

It is useful for:

Building networks
Configuring devices
Testing connectivity
Performing normal network operations

Simulation Mode

Simulation mode allows students to observe network communication step-by-step.

It can be used to:

- Observe packets moving through a network.
- See which devices process a packet.
- Examine protocols involved in communication.
- Understand how data travels through a network.

Locate the Realtime and Simulation Controls in Packet Tracer

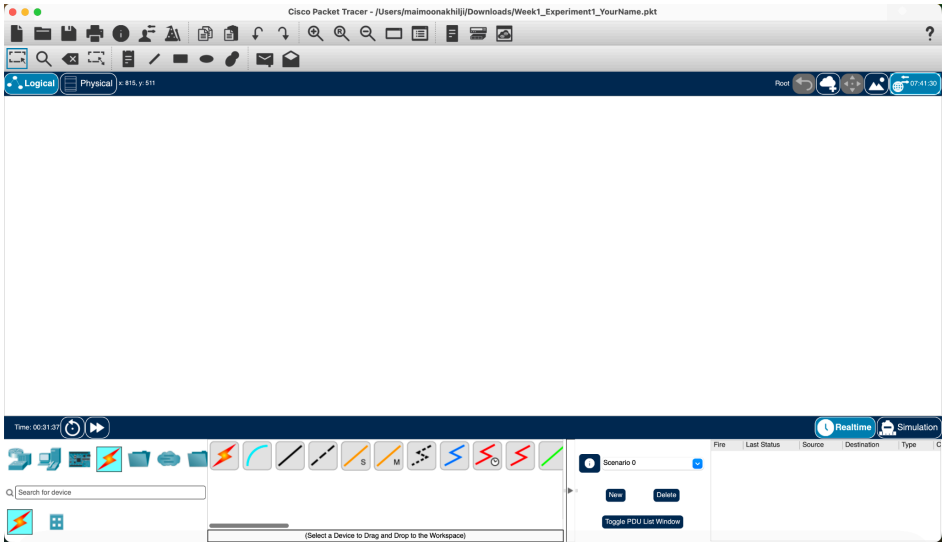
At the bottom-right area of the Packet Tracer window, locate:

- Realtime
- Simulation



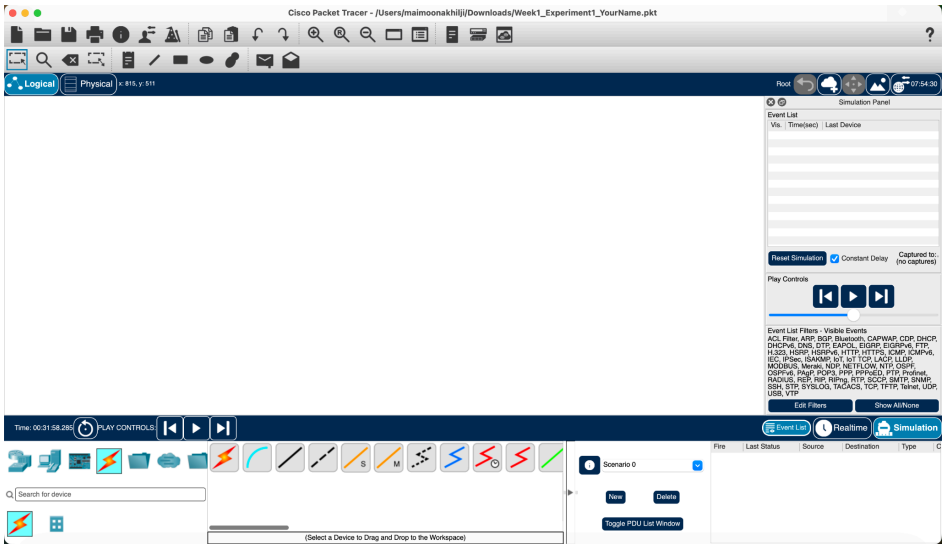
1. **Select Realtime Mode**

   Click **Realtime**. This is the default mode when Packet Tracer starts.



2. Select Simulation Mode

Click **Simulation**. The Packet Tracer interface will change to provide controls for observing network events.



3. Return to Realtime Mode

Click **Realtime** again. Students should be able to switch between both modes whenever required.

Basic PC–Switch–PC Network

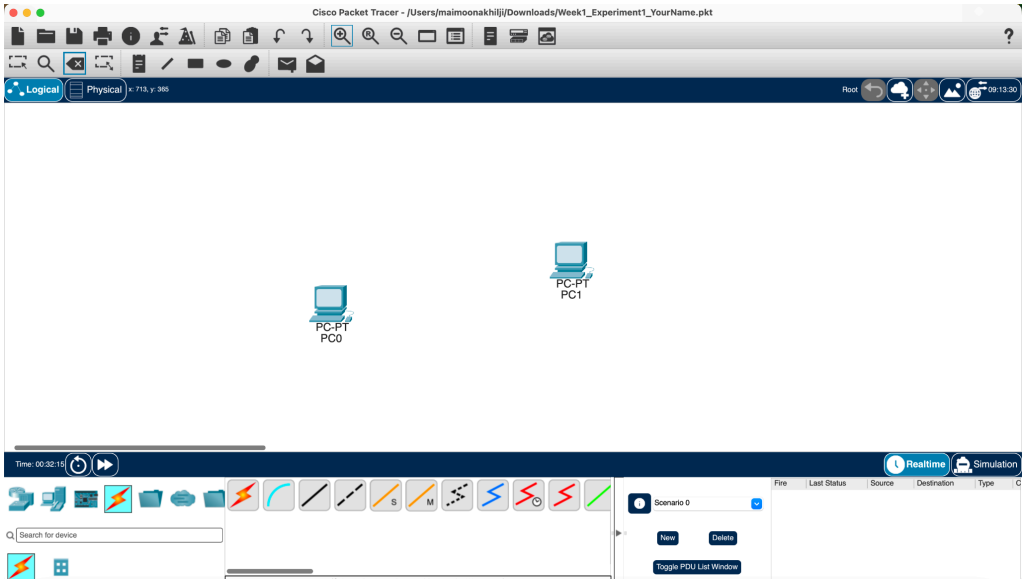
Objective

By completing this experiment, students will be able to:

- Place PCs and a switch in Packet Tracer.
- Connect devices using a network cable.
- Configure IPv4 addresses on PCs.
- Test connectivity using the ping command.
- Observe communication in Realtime and Simulation modes.

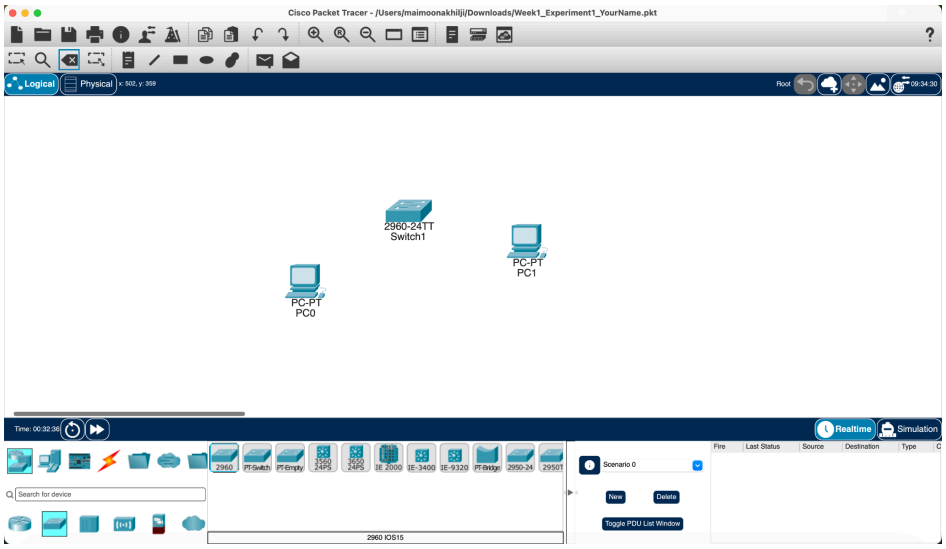
Step 1: Place Two PCs

- From **End Devices**, select **PC**.
- Place two PCs on the logical workspace.
- Packet Tracer will normally name them as PC0 and PC1.
- Arrange them with some space between them.



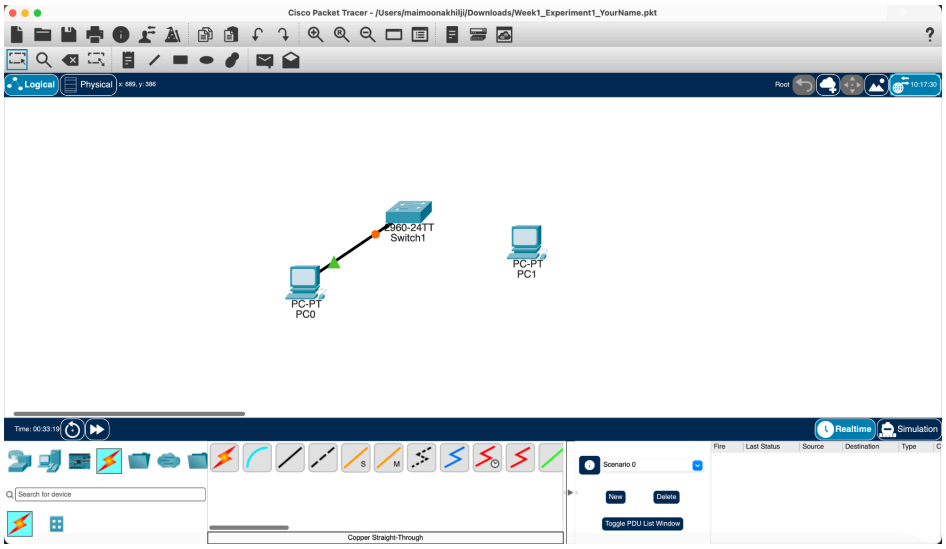
Step 2: Place a Switch

- Select: **Network Devices → Switches**
- Choose a basic switch, such as **2960**, and place it between the two PCs. The topology should look approximately like: PC0 -------- Switch0 -------- PC1



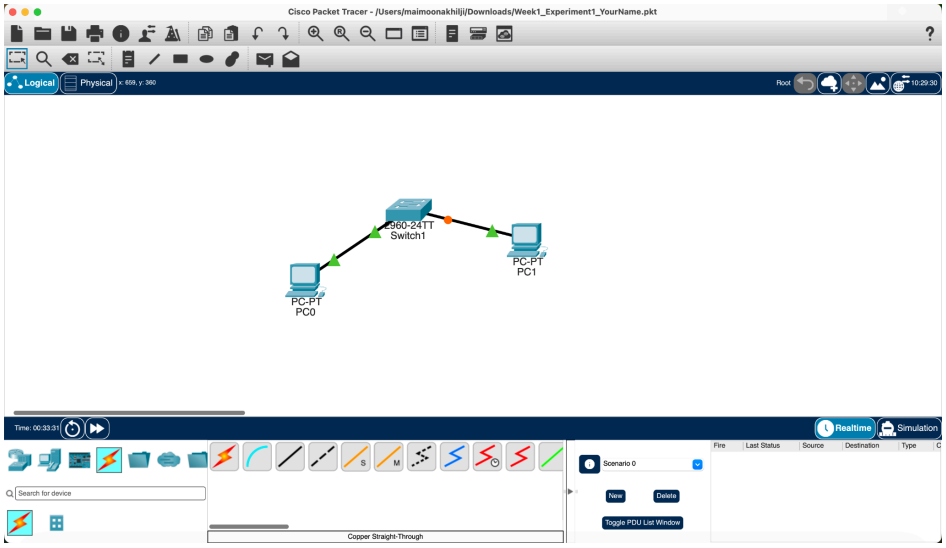
Step 3: Connect PC0 to the Switch

- Select **Connections**.
- Choose: **Copper Straight-Through**
- Click **PC0** and select its **FastEthernet** interface.
- Then click **Switch0** and select an available **FastEthernet** port, such as: FastEthernet0/1



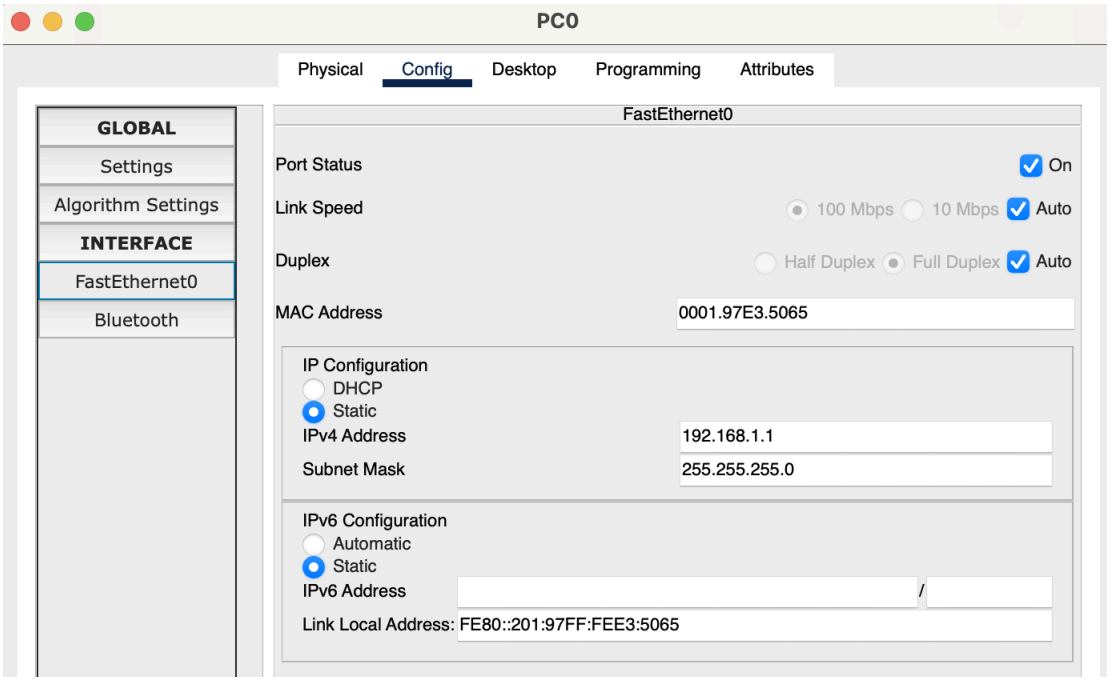
Step 4: Connect PC1 to the Switch

- Again select **Copper Straight-Through**.
- Connect: **PC1 FastEthernet0 → Switch0 FastEthernet0/2**
- The completed topology should look like: PC0 -------- Switch0 -------- PC1



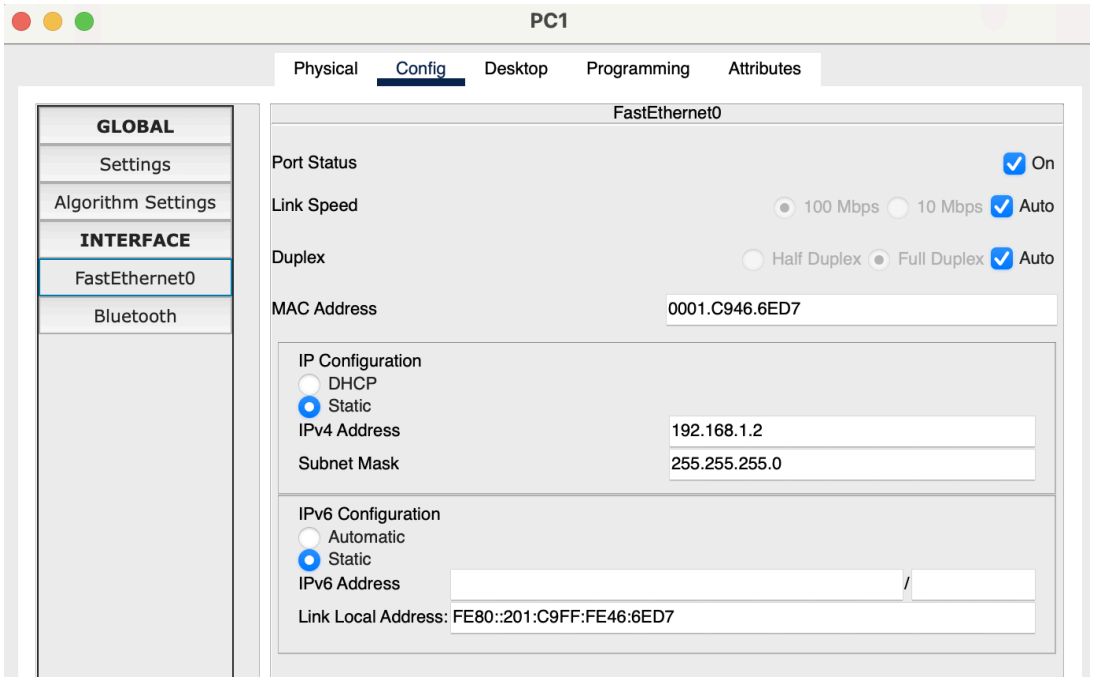
Step 5: Configure the IP Address of PC0

- Click **PC0**.
- Select: **Desktop → Fast Ethernet → IP Configuration**
- Enter:
  - **IP Address:** 192.168.1.1
  - **Subnet Mask:** 255.255.255.0
- Leave the default gateway empty.
- Close the configuration window.



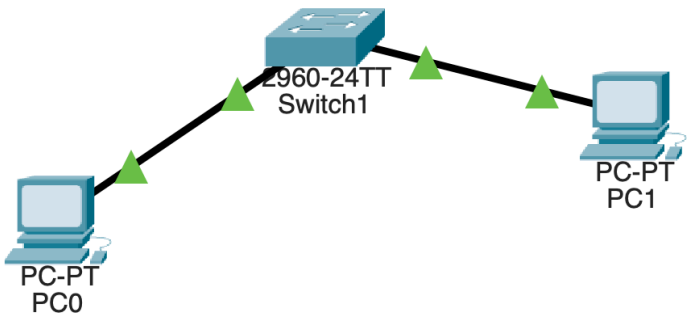
Step 6: Configure the IP Address of PC1

- Click **PC1**.
- Go to: **Desktop → Fast Ethernet → IP Configuration**
- Enter:
    - **IP Address:** 192.168.1.2
    - **Subnet Mask:** 255.255.255.0
- Leave the default gateway empty.



Step 7: Wait for the Links to Become Active

- Observe the connections between the PCs and switch.
- The link indicators should eventually become **green**, indicating that the connections are active.
- If the links are not active immediately, wait a few seconds.

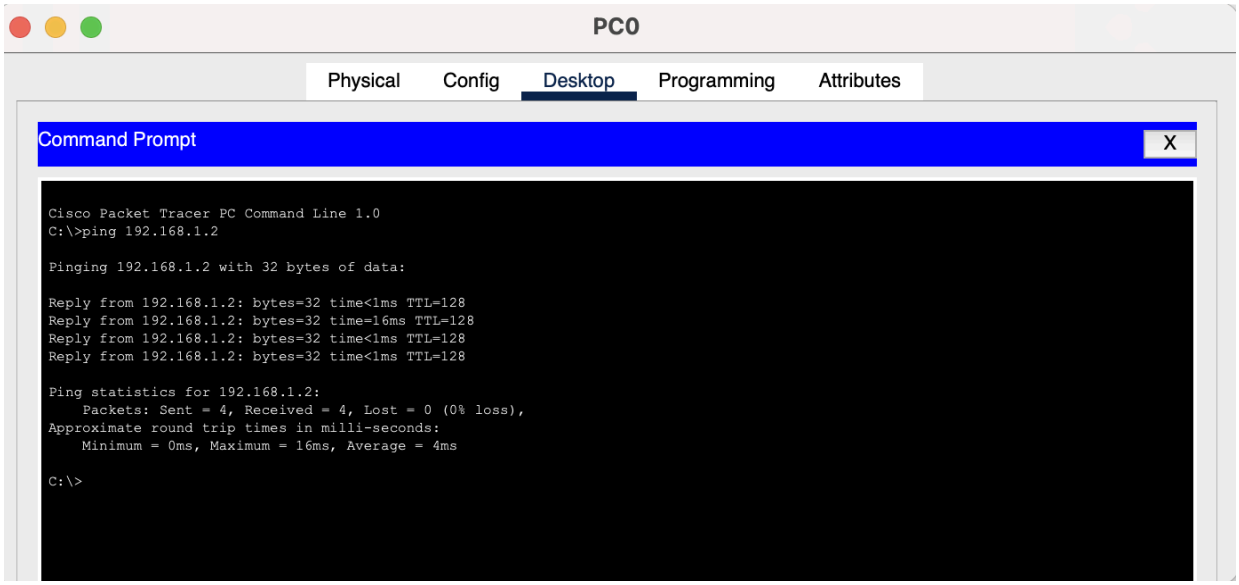


Step 8: Test Connectivity from PC0

- Open **PC0**.
- Go to: **Desktop → Command Prompt**
- Type: ping 192.168.1.2
- Press **Enter**.
- If the configuration is correct, PC0 should receive replies from PC1.

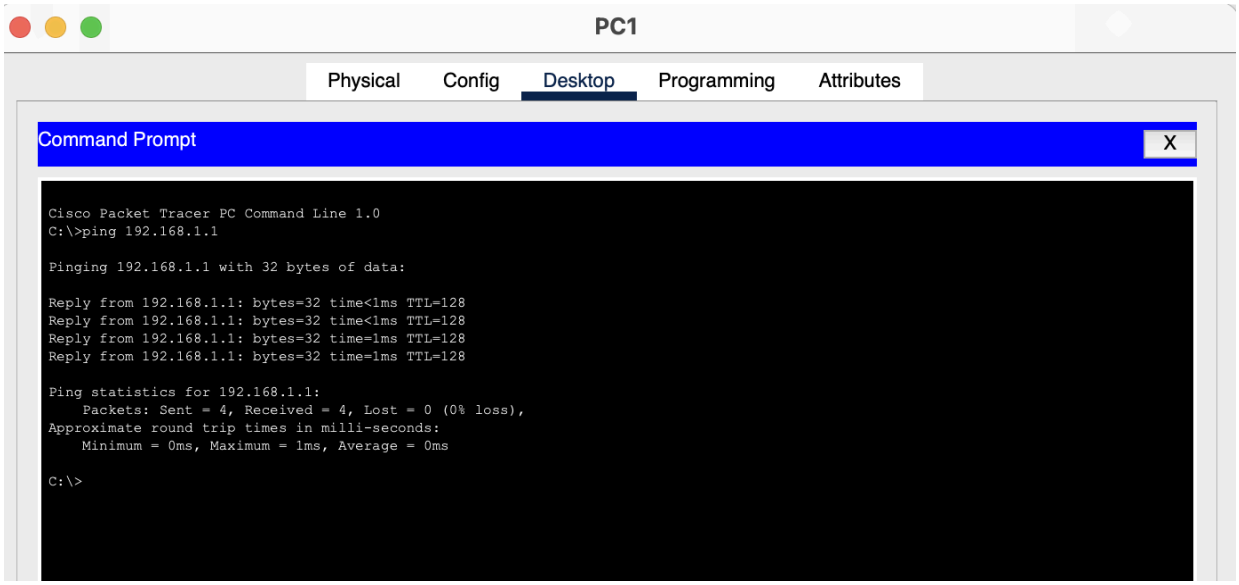
Expected Result:

- Reply from 192.168.1.2
- The exact Packet Tracer output may contain additional information such as bytes, time, and TTL.



Step 9: Test Connectivity from PC1

- Open **PC1 → Desktop → Command Prompt**.
- Enter: ping 192.168.1.1
- Press **Enter**.
- PC1 should receive replies from PC0.

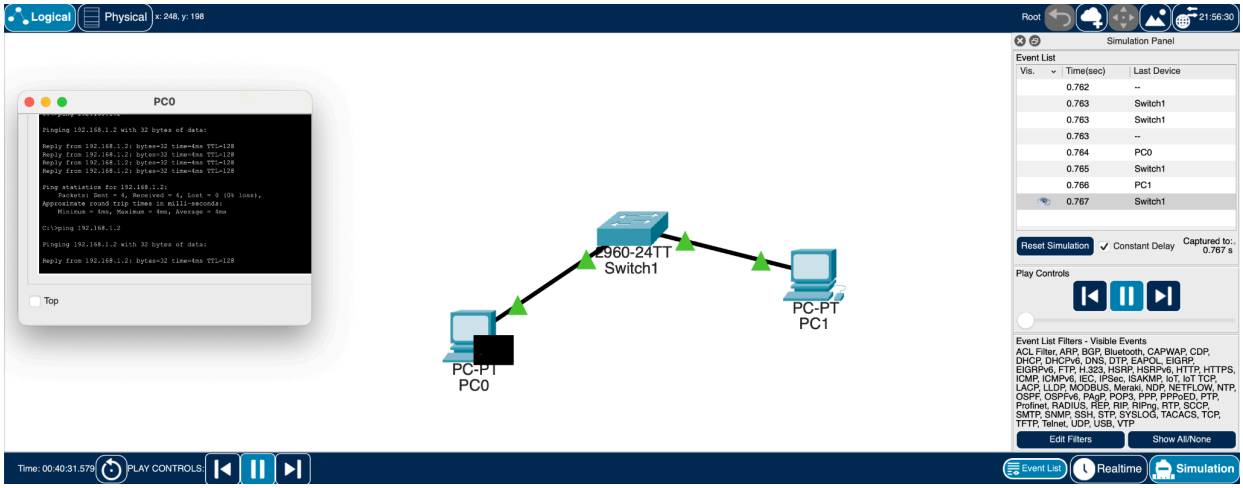


Step 10: Observe the Network in Simulation Mode

- Switch Packet Tracer from **Realtime** to **Simulation** mode.
- Open PC0's command prompt and send: **ping 192.168.1.2**
- Observe the packets appearing in the simulation panel. Use the available controls to move through the communication events step-by-step.

Step 11: Observe Packet Movement

- Use the simulation controls to observe how the packet travels:
  - PC0 → Switch0 → PC1
- Then observe the response:
  - PC1 → Switch0 → PC0
- Pay attention to the devices through which the packet travels.



Step 12: Return to Realtime Mode

- After completing the observation, switch back to: **Realtime**
- The network should continue to operate normally.

Verification Checklist

Verify the following:

Check	Expected Result
PC0 connected to Switch0	Connected
PC1 connected to Switch0	Connected
PC0 IP address	192.168.1.1
PC1 IP address	192.168.1.2
Subnet mask on both PCs	255.255.255.0
PC0 → PC1 ping	Successful
PC1 → PC0 ping	Successful
Realtime mode	Working
Simulation mode	Working
Packet movement observed	Yes

Common Troubleshooting

Problem 1: Ping fails

Check:

- IP addresses
- Subnet masks
- Cable connections
- Whether both PC links are green

Problem 2: Link is red

- Check whether the correct cable were selected.
- Wait a few seconds after connecting the devices.

Problem 3: Wrong IP address

- Open:
  - **PC → Desktop → IP Configuration**
- and verify the entered values.

Problem 4: Simulation shows too many events

- Use the simulation controls and filters to focus on the required communication or Reset.

Recap

1. **What is Cisco Packet Tracer?**
   Cisco Packet Tracer is a network simulation tool used to create, configure, and test computer networks virtually.

2. **What is the purpose of the Logical Workspace?**
   The Logical Workspace is used to build and view the network topology by placing and connecting devices.

3. **What is the difference between Realtime and Simulation mode?**
   Realtime mode shows normal network operation, while Simulation mode allows us to observe packets moving through the network step by step.

4. **Which device connects the two PCs in this experiment?**
   A network switch connects the two PCs.

5. **Which cable was used to connect the PCs to the switch?**
   A Copper Straight-Through cable was used.

6. **What IP address was assigned to PC0?**
   PC0 was assigned the IP address **192.168.1.1**.

7. **What IP address was assigned to PC1?**
   PC1 was assigned the IP address **192.168.1.2**.

8. **Why were both PCs given the same subnet mask?**
   Both PCs use the same subnet mask because they belong to the same network.

9. **Which command was used to test connectivity?**
   The `ping` command was used to test connectivity.

10. **What path did the packet follow when PC0 communicated with PC1?**
    The packet followed the path **PC0 → Switch0 → PC1**.
